

UNIT 5

Cloud Service Models and Cloud Infrastructure**5.1. Jericho Cloud Cube Model:**

- Cloud computing offers a huge possibility for scalability, at almost instantaneous availability and low cost.
- Business managers requires IT operations to assess the risks and benefit this representation of computing model.
- The Jericho forum is an independent group of international information security leaders, have added their input as to how to collude securely in the clouds.
- The Jericho Cloud Cube Model portrays the multidimensional elements of cloud computing, that frames not only cloud use cases but also how they are set up and used.

Objectives of Jericho Forum:

The Jericho Forum's objectives associated to cloud computing are unique – “enabling secure combination in the appropriate cloud formations suited best to the business needs”.

The Jericho forum: -

- ❖ points out that in clouds not everything is best implemented; it may be best to conduct some business functions using a conventional non-cloud approach
- ❖ Explains the Jericho forum identified different cloud formations.
- ❖ describes benefits, key characteristics, and risks of each cloud formation
- ❖ Provide a framework for seeking in more detail the nature of different cloud formations and the issues that demands to answer to make secure places to work in and make them safe.

The Jericho Forum is actively supporting solution providers and merchants to develop the missing capabilities and services to assure customers are protected from the rough association of clouds.

Protecting our Data

First, it is necessary to categorize our data so as to know what rules must be applied to protecting it:

- Its sensitivity - must it exclusively exist at specific trust levels? If so, which?
- What supervisory /compliance restrictions apply – e.g. Must it remain within your national borders?

We only can meet this requirement if we have comprehensively adopted standards for:

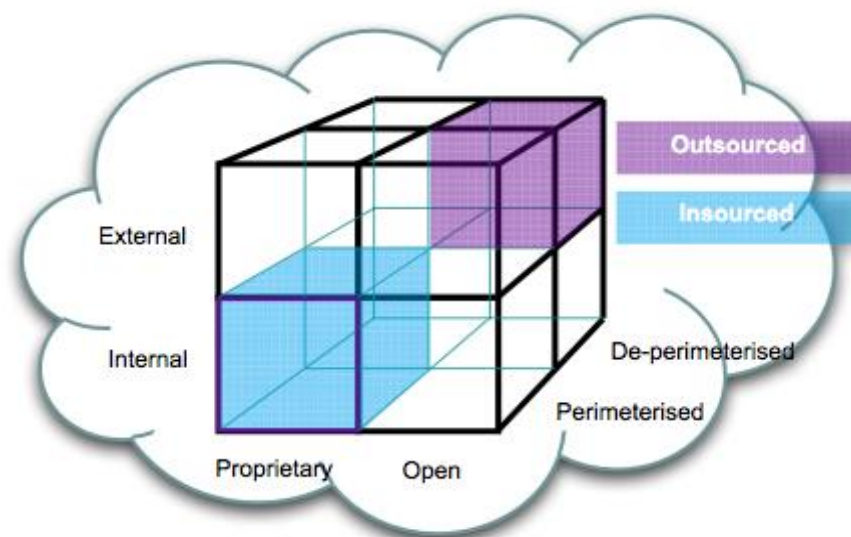
- a data classification model that is easy enough for all originators of data to use – for eg the G8 Traffic Light Protocol

- an associated basis for managing trust levels
- Regulated metadata that signals to “cloud security” what security must be applied to each item of data.

With consideration of what security, we must apply to our data, we’re in a position to take decision:

- what data and processes to migrated to the Clouds?
- At what level we want to perform in the Clouds? Cloud models isolate layers of business service from one another, for example, Infrastructure, Process, Platform, Software, and Process.
- Which Cloud composition are best suited to our need?

The Cloud Cube Model:



The Cloud Cube Model

The Jericho forum has identified 4 gauge for judgment to differentiate cloud formations from each other and manner of their provisions.

The cloud cube model summarizes these four dimensions.

Cloud Cube Model Dimensions

1. Dimension: Internal (I) / External (E)
2. Dimension: Proprietary (P) / Open (O)
3. Dimension: Perimeterised (Per) / De-perimeterised (D-p) Architectures
4. Dimension: Insourced / Outsourced

1. Dimension: Internal (I) / External (E):

- This dimension defines the physical location of the data such as where does the cloud form, we want to use lie, inside or outside your organization's boundaries.
- It is Internal If it is within your own physical boundary.
- It is External If it is not within your own physical boundary.
- For example, while Amazon SC33 would be external at some location "off-site, virtualized hard disks in an organization's data center will be internal.

2. Dimension: Proprietary (P) / Open (O):

- This is the dimension that represents the state of ownership of the cloud technology, interfaces, services, etc.
- It indicates the degree of interoperability, allowing "data/application transportability" between other cloud forms and your own systems, and the ability to pull out your data from a cloud form or to migrate it to another without force.
- **Proprietary** means that the organization that provides the service is keeping the means of arrangement under their ownership. As a result, when functioning in clouds that are proprietary, you may not be allowed to move to another cloud provider without specific effort or investment. Mostly the more innovative technology progress occurs in the proprietary realm.
- • **Open** clouds use technology that is not proprietary, meaning that there are likely to be more suppliers, and user are not as strained in being able to share your data and using the same open technology collide with selected parties. Open services tend to be those that are computerized and widespread, and apparently most likely a published open standard, for example, email (SMTP).

3. Dimension: Perimeterised (Per) / De-perimeterised (D-p) Architectures:

- Architectural mindset is represented in the third dimension.
- **Perimeterised** indicates continuing to operate within the classical IT perimeter, often indicated by "network firewalls". This approach discourages collaboration. When operating in the parameterised areas, you may simply prolong your own organization's perimeter using a VPN and operating the virtual server in your own IP domain into the external cloud computing domain, using your own directory services for access control. Then, when the computing task is finished you can pull out your perimeter back to its original classical position.
- **De-perimeterised**, considers that the system perimeter is architected succeeding the principles put forward in the Jericho Forum's Commandments and collusion Oriented Architectures Framework. In a de-perimeterised frame the data would be wrapped with meta-data and mechanisms that from inappropriate usage would protect the data .In a de-perimeterised environment an organization can associate securely with selected parties globally over COA capable network.

4. Dimension: Insourced / Outsourced:

- We define a 4th dimension having 2 states in every one of the 8 cloud forms: Per(IP,IO,EP,EO) and D-p(IP,IO,EP,EO), that answers to the question “Who do you want running our Clouds?”:
- **Outsourced:** In outsourced dimension, the service is provided by a 3rd party.
- **Insourced:** In insourced dimension the service is provided by your own staff under your control
- These 2 states express who is managing the delivery of the cloud service(s) that you are using.
- This is generally a policy issue (i.e. not a technical or architectural decision but a business decision) which must be represented in a contract with the cloud service provider

5.2. Introduction to Cloud Computing Services:

1. SaaS
2. PaaS
3. IaaS/HaaS

Already described in Unit one in detail ,Please find out the benefits of each services along with detail explanation of CAAS and DAAS and NAAS here in this unit.

4. CaaS
- 5.DBaaS

1. SaaS (Software as a Service):**Benefits:**

- Cost-Effective: Pay only for what you use.
- Reduced time: Users can run most SaaS apps directly from their web browser without needing to download and install any software.
- Accessibility: We can Access app data from anywhere.
- Automatic updates: Rather than purchasing new software, customers rely on a SaaS provider to automatically perform the updates.
- Scalability: It allows the users to access the services and features on-demand.

2. PaaS:(Platform as a Service):**Benefits:**

- Simple and convenient for users: It provides much of the infrastructure and other IT services, which users can access anywhere via a web browser.
- Cost-Effective: It charges for the services provided on a per-use basis thus eliminating the expenses one may have for on-premises hardware and software.
- Efficiently managing the lifecycle: It is designed to support the complete web application lifecycle: building, testing, deploying, managing, and updating.
- Efficiency: It allows higher-level programming that reduces complexity.

3. IaaS (Infrastructure as a Service):**Benefits:**

- Cost-Effective: Eliminates capital expense and reduces ongoing cost and IaaS customers pay on a per-user basis, typically by the hour, week, or month.
- Website hosting: Running websites using IaaS can be less expensive than traditional web hosting.
- Security: The IaaS Cloud Provider may provide better security than your existing software.
- Maintenance: There is no need to manage the underlying data center or the introduction of new releases of the development or underlying software. This is all handled by the IaaS Cloud Provider.

4. CaaS (Communication as a Service):

- Communications as a Service (CaaS) is an outsourced enterprise communications solution that can be leased from a single vendor.
- Such communications can include voice over IP (VoIP or Internet telephony), instant messaging (IM), collaboration and videoconference applications using fixed and mobile devices.
- The CaaS vendor is responsible for all hardware and software management and offers guaranteed Quality of Service (QoS).
- CaaS allows businesses to selectively deploy communications devices and modes on a pay-as-you-go, as-needed basis.
- This approach eliminates the large capital investment and ongoing overhead for a system whose capacity may often exceed or fall short of current demand.

Features:

- Integrated and Unified Communication.
- No Investment Required.
- Flexibility & Scalability.
- No Risk of Obsolescence.
- No Maintenance Cost Incurred.
- Ensure Business Continuity.

Benefits:

- CaaS provides an economical way to deliver communication service to its customer
- CaaS vendor provides 24/7 service to its customers.
- Customer receiving services from CaaS vendor do not have to indulge and invest in managing the components of CaaS.
- CaaS vendor offers flexible service as they charge according to pay as you go basis.
- CaaS provide scalable services as they provide service based on customer's demand.

5. DBaaS (Database as a Service):

- DBaaS(Database as a service) is a cloud computing managed service offering model that enables users to set up, operate, manage and scale with some form of access to a Database without the need for setting it up on physical hardware, installing software, or configuring it for performance, Database management by themselves.

Popular DBMS offerings include:

- Amazon
- Google Cloud
- Microsoft Azure
- Compose (IBM)
- IBM Cloudant
- MongoDB Atlas
- Oracle Database Cloud Service

Features:

- Self-service: DBaaS allows the provision of Databases effortlessly to Database consumers from various backgrounds and IT experience.
- On-demand - While generating overall IT savings, You pay for what you use.
- Dynamic: Based on the resources available, it delivers a flexible Database platform that tailors itself to the environment's current needs.
- Security - A team of experts at your disposal, continuously monitoring your Databases.
- Automation: Automates Database administration and monitoring.

Benefits:

- It simplifies the process of requesting and deploying VM's, as well as the final provisioning of Database servers, instances, and Databases with self-provisioning
- When a remote service provider hosts the cloud Database, businesses' need to incur expensive hardware or software licenses is ruled out.
- It is the responsibility of the Database provider to manage and maintain Database hardware and software.
- The hefty power bills for ventilation and cooling bills to keep the servers running are eliminated.
- Human resources required to manage the maintenance of the system is eliminated.
- On the other hand, workforce or human resources saved can be utilized for the company's different activities like running and growth of its business.
- Since DBaaS is hosted off-site, the organization is free from the hassles of power or network failure or disruption issues.

5.3. Cloud Computing at Amazon:(AWS):

- Cloud computing is the on-demand delivery of IT resources over the Internet with pay-as-you-go pricing. Instead of buying, owning, and maintaining physical data centers and servers, you can access technology services, such as computing power, storage, and databases, on an as-needed basis from a cloud provider like Amazon Web Services (AWS).
- AWS is an online platform that provides scalable and cost-effective cloud computing solutions.
- AWS is a broadly adopted cloud platform that offers several on-demand operations like compute power, database storage, content delivery, etc., to help corporates scale and grow.

History of AWS

In the year 2002 - AWS services were launched

In the year 2006- AWS cloud products were launched

In the year 2012 - AWS had its first customer event

In the year 2015- AWS achieved \$4.6 billion

In the year 2016- Surpassed the \$10 billion revenue target

In the year 2016- AWS snowball and AWS snowmobile were launched

In the year 2019- Released approximately 100 cloud services

Applications of AWS

Here are some of the most common applications of AWS:

1. Storage and Backup

One of the reasons why many businesses use AWS is because it offers multiple types of storage to choose from and is easily accessible as well. It can be used for storage and file indexing as well as to run critical business applications.

2. Websites

Businesses can host their websites on the AWS cloud, similar to other web applications.

3. Gaming

There is a lot of computing power needed to run gaming applications. AWS makes it easier to provide the best online gaming experience to gamers across the world.

4. Mobile, Web and Social Applications

A feature that separates AWS from other cloud services is its capability to launch and scale mobile, e-commerce, and SaaS applications. API-driven code on AWS can enable companies to build uncompromisingly scalable applications without requiring any OS and other systems. Companies using AWS

List of companies using AWS:

Netflix

Intuit

Coinbase

Finra

Johnson & Johnson

Capital One

Adobe

Airbnb

AOL

Hitachi

Advantages of AWS Services**1. Security**

There is a false misconception that data stored in a public cloud is not secure. On the contrary, not only does AWS offer security tools that are cheaper than other alternatives, but it is one of the most secure, extensive, and reliable cloud platforms.

2. Global Availability

AWS has 80 zones across 25 geographic regions global data centers.

2. Scalability and Flexibility

AWS offers unlimited flexibility and scalability on demand. This enables organizations to plan their infrastructure roadmap on a subscription basis without full commitment.

3. Little Investment

AWS cloud services enable companies to save expenditures on extra software and hardware. There is no physical data required, which ultimately lowers down operating costs.

AWS Services

Amazon has a list of services:

Compute service

Storage

Database

Networking and delivery of content

Security tools

Developer tools

Management tools

IoT

Security

Gaming's

Enterprise applications

5.4. Cloud Computing from the Google Perspective:

- Google Cloud Platform is a set of Computing, Networking, Storage, Big Data, Machine Learning and Management services provided by Google that runs on the same Cloud infrastructure that Google uses internally for its end-user products, such as Google Search, Gmail, Google Photos and YouTube.

Google focuses on three things

- a. Making cloud more accessible
- b. Keep connectivity pervasive
- c. Make the client more powerful



Google offers a wide range of Services.:

- Compute
- Networking
- Storage and Databases
- Big Data
- Machine Learning
- Identity & Security
- Management and Developer Tools

Advantages of GCP:

- Better Pricing Than Competitors
- Private Global Fiber Network
- Live Migration of Virtual Machines
- Improved Performance
- State of the Art Security
- Dedication to Continued Expansion
- Redundant Backups

Top Companies in GCP:

- Verizon: \$79.9 million
- Twitch Interactive: \$63.8 million
- CenturyLink: \$61.8 million
- LinkedIn Corp: \$41.1 million
- NewsCorp: \$40.9 million
- SAP: \$40.1 million
- Facebook, Inc: \$ 39.4 million
- Intel: \$39.2 million
- Yahoo: \$46.6 million
- Marriott International: \$35.9 million

5.5. Microsoft Azure:

- Windows Azure, which was later renamed as Microsoft Azure in 2014, is a cloud computing platform, designed by Microsoft to successfully build, deploy, and manage applications and services through a global network of data centers.
- Azure uses large-scale virtualization at Microsoft data centers worldwide and it offers more than 600 services.

Azure Services:

1. Azure Application Services

- Azure application services help you develop, deploy, and maintain various applications on the Azure platform.
- These include Azure AI, Azure Analytics, Azure IoT, Azure Active Directory (AD), Azure Media Services, and Azure Scheduler.

2. Azure Data Services

- Azure data services store and manage data on cloud. Microsoft Azure comes with a range of data services: Azure Storage, Azure SQL Database, Azure DocumentDB, Azure StorSimple, and Azure Redis Cache.

3. Azure Development Services

- Microsoft Azure offers multiple development tools and services to enhance the overall software development and deployment process.
- Here's a quick look at one of the essential services by Microsoft i.e. Azure DevOps, enabling continuous delivery to customers.

4. Azure Compute Services

- Azure compute services are the hosting services responsible for hosting and running the application workloads.
- These include Azure Virtual Machines (VMs), Azure Container Service, Azure App Services, Azure Batch, and Azure Service Fabric.

5. Azure Network Services

- Azure Network Services refer to those services that perform networking operations within Azure and between Azure and on-premise infrastructure.
- These include Azure Virtual Network, Azure ExpressRoute, Azure-provided DNS, Azure Traffic Manager and Azure Content Delivery Network (CDN).

The detail comparison of these 3-cloud service provider is given in separate pdf.

Open-Source Software Platforms for Private Clouds:**1. OpenShift:**

- ✓ OpenShift is Red Hat's Platform-as-a-Service that allows developers to quickly develop, host, and scale applications in a cloud environment.
- ✓ With OpenShift, businesses have a choice of offerings, including online, on-premise, and open-source project options.

2. Cloudify:

- ✓ Cloudify is provided as an opensource framework, however unlike how Openshift focuses on application stacks and architectures, Cloudify is designed with the developer in mind, and aims to make it easier to deploy new, simple applications.

3. Tsuru:

- ✓ Tsuru is a platform as a service product from Globo.com.
- ✓ The software that runs the service is open-sourced under the Tsuru name, and is available on GitHub.
- ✓ Developers can use Tsuru to deploy web applications in different languages on the platform.
- ✓ "Tsuru supports most web applications, so long as they can run on Linux. Tsuru takes care of maintaining the services underlying the application, high availability, scaling and healing the application as needed."

4. Cloud Foundry:

- ✓ Developed by VMware, Cloud Foundry provides dynamic products and as a platform as a service as part of Pivotal software.
- ✓ Their motto, "Built BY industry leaders, FOR industry leaders", is telling of its large open community of Ruby developers and users who support open source services. Cloud Foundry describes the products features:
- ✓ "Among the services Cloud Foundry offers for its hosted solution are MySQL DB, VFabric Postgres, MongoDB, Redis, and RabbitMQ."

5. WSO2 Stratos:

- ✓ WSO2 Stratos supports more core services than other available PaaS options today and is a good option for enterprises that seek to "extend the flexibility and innovation achieved from implementing heterogeneous environments on-premises, into the cloud,"
- ✓ "WSO2 Stratos offers extensible cartridge architecture, enhanced cloud deployment support for multiple IaaS, as well as easy SaaS app development."

6. Alibaba**7. Stackato**